School of Computing, Engineering, and Physical Sciences
Assessment Guidance Coversheet

Module Code:	COMP10013	
Module Title:	Dynamic Web Technologies	
Module Co-ordinator:	Dr Pablo Salva Garcia	
Assessment set by:	Dr Pablo Salva Garcia	
Learning Outcomes Assessed:	**LO1**: Demonstrate a deep and thorough understanding of the use of client-side frameworks and HTML5/Native APIs for the development of dynamic web sites and mobile apps. **LO2**: Apply client-side frameworks and HTML5 in real web and mobile app applications. **LO2**: Evaluate the advantages and disadvantages of using web and mobile app frameworks.	
Issued:	14/03/2026	
Deadline:	14/04/2026	
Feedback[1]:	28/04/2026	
Individual / Group assessment:	Individual ☒	Group ☐

Anonymity
The University Regulations makes it clear that all assessments are marked anonymously (Regulation 3.4) unless the assessment itself renders anonymity impossible e.g. placements, presentations, practical assessments. Unless otherwise stated, do not put your name or Banner ID on your submission.

Referencing
The standard referencing style at UWS is **Cite Them Right (CTR) Harvard** and your references should be in accordance with these guidelines. The University provides a drop-down menu that lets you see examples of how you should reference journals, books, websites etc. Guidance on referencing styles can be found on the UWS Library website: https://uws-uk.libguides.com/referencing

Extenuating Circumstances (ECS)
The University recognises that, from time to time, you may encounter issues which may prevent you from being able to submit or undertake an assessment. Where this is the case, you can complete an Extenuating Circumstances Submission (ECS) for consideration. The ECS will be forwarded to the School Assessment Board to take account of this declaration in recording your module marks. Guidance on ECS claims can be found: https://www.uws.ac.uk/current-students/supporting-your-studies/exams-assessment-appeals/academic-appeals-extenuating-circumstances/

Generative AI

[1] Note that any mark / grade you receive is provisional and may be subject to change until there has been internal moderation, external examination and ratification at the School Assessment Board.

The type of Generative AI you are allowed to use within your assessment is:

Type	Description	Allowed
1	**Restricted** Only the use of routine and established tools, such as auto-transcription, spell checkers, grammar check is permitted.	☒
2	**Specified** Generative AI may be used for clearly delineated tasks as appropriate / allowed / recommended, although its use is **not** mandatory in order to complete the assessment.	☐
3	**Open** No specific restrictions but with requirement to track key stages / tools utilised. The use of such tools is **not** mandatory in order to complete the assessment. This may include: (i) Socratic chatbot, (ii) summarisation, simplification, synthesis, and translation, (iii) generative illustrative media content, and (iv) support production of multimedia artefacts.	☐
4	**Embedded** Generative AI is a feature of the assessment itself. Here the use of Generative AI is a focal aspect of the assessment. This may include (i) using named tools for specific outcome, (ii) output comparison and critique, and (iii) develop or error check code.	☐

Assessment Student Declaration Coversheet

All students are expected to complete the Student Declaration and insert that as the first page of your submission. The Module Co-ordinator will have uploaded this to Aula for you to complete.

- **For written assessments, insert the Declaration as the first page of your document**.
- For assessments not in the written format (e.g. video, audio, presentation, or practical work), submit the Declaration as a separate file.
- For group assessments, unless otherwise directed by your lecturer, the group may submit a single shared declaration.

Assessment guidance

Course Work 01 (CW01) – 100% of the total Mark

Note: Provisional and subject to change. If any changes occur, you will be notified accordingly.

Submission Overview:

This assignment consists of two parts: A Formal Documentation part submitted in written format (Part A) and a Practical Work part (Part B) assessed via a demonstration.

The assignment parts are as follows:

• Part A - 40%: A written report documenting your work.

• Part B - 60%: A technical demonstration of your practical work assessed during a scheduled presentation.

This assessment makes up 100% of the module mark. Per the assessment regulations, the pass mark for undergraduate students is 40%.

Submission Instructions: Part A - Written Component - Formal Documentation (40%)

The written component must be submitted via the Aula (Turnitin) assignment box titled "*Part A: Written Component*" and must include:

- Your Student Information: Include vital details such as your name, Banner ID, and any necessary usernames/passwords that an examiner might need to access or evaluate your work.
- Link to Your GitHub Repository: Provide the link to your project's GitHub repository containing the source code. Please note that commits made to the repository after the submission deadline will not be considered for assessment. Ensure your README.md file has comprehensive instructions to clone, install, and run your application.
- Content: You should document the development process and include the following: a clearly defined topic or feature covered during this module; an introduction to your project explaining how the chosen technical feature contributes to its successful completion and performance; a breakdown of your system architecture and functionality, supported by reproducible examples to enhance understanding; a step-by-step tutorial guiding the reader on how to use your project; and a conclusion with overall recommendations, summarising lessons learned and providing actionable insights.

Format requirements: Part A - Written Component - Formal Documentation (40%)

1. Your submission can be either in a word .docx file or a PDF.
2. Your submission file name must follow this format: ***DWT-CW01-T2-Attempt-B123456-Your-Name.docx***
    i. **Replace B123456 with your matriculation number (banner ID) and include your full name instead of "Your-Name".*

The report should be between **10 and 15 pages** in length. If you require additional pages, please consult your lecturer. (+- 10% is allowed without previous consultation.)

Submission Instructions: Part B - Practical Component – Technical Demonstration (60%)

Demonstrations offer you the opportunity to present your practical work and communicate your knowledge. While it's ideal for your practical work to be available on a publicly accessible URL (24/7 operation not required), hosting on a local server or other non-public URLs is also acceptable during the demonstration. However, your code must be 24/7 available on a git repository. Demonstrations will be in-person at office **E357**, on-line presentations will be

considered for exceptional circumstances but note presenting a mobile phone application through the webcam might be challenging. Video recordings are not allowed. The presentations will last approximately 15 to 20 minutes. Please ensure your mobile application is ready for presentation before your demonstration.

Plagiarism:

Plagiarism involves gaining an unfair advantage in a university assessment by using another person's work or accessing unauthorised materials. Examples include:

Plagiarism is a serious offence and is considered fraud by the university. Any instances will be investigated and handled appropriately.

How to Avoid Plagiarism:

For more information on avoiding plagiarism and proper referencing, visit the Centre for Learning and Teaching's Study Skills Guide:

http://www.uws.ac.uk/clt/students/studyskills/guides/plagiarism.htm

Assessment Breakdown

Note: Provisional and subject to change. If any changes occur, you will be notified accordingly.

Please refer to the following pages for a detailed description of the coursework, along with the marking rubric. A template has also been provided on Aula to help you structure the document you are required to submit.

Assessment Breakdown: Part A - Written Component - Formal Documentation (40%)

This assessment evaluates your ability to **clearly document, explain, and critically reflect on the development of a React-based mobile application** (Progressive Web App or React Native). The written submission should demonstrate not only *what* you built, but *why* and *how* it was designed, implemented, and used.

Your report will be assessed using the criteria below, which directly correspond to the marking rubric.

1. **Presentation & Visual Appeal (Max. 5 Marks)**

Your report should be professionally presented and easy to read. This includes:

- Clear structure with appropriate headings and subheadings.
- Consistent formatting, layout, and visual hierarchy.
- Effective use of diagrams, screenshots, or figures where appropriate to support explanations.

2. Introduction & Feature Explanation (Max. 10 Marks)

You must clearly introduce your project and the **key technical feature or topic(s)** you have chosen from the module content.

This section should include:

- A concise introduction to the application and its purpose.
- Identification of **one or more specific React or mobile development features** covered in the module (e.g. React Hooks, PWA features, CRUD operations, storage solutions, hardware APIs, Expo, APK generation, or a pre-approved alternative).

- An explanation of how the chosen feature(s) contribute to the functionality, performance, or user experience of the application.
- Practical examples illustrating how and where the feature is used within your project.

3. Technical Understanding & Documentation (Max. 10 Marks)

This section assesses your **depth of technical understanding** and ability to explain implementation decisions.

You should:

- Describe the system architecture and main components of your application.
- Explain how the selected feature(s) have been implemented, supported by relevant code snippets or pseudo-code where appropriate.
- Demonstrate understanding of how different parts of the application interact.
- Use clear, reproducible examples to support your explanations.

The emphasis is on **explaining design and implementation choices**, not simply listing technologies.

4. Project Setup & User Guide (Max. 10 Marks)

You must provide a **clear, step-by-step guide** that enables another user or developer to install and use your application.

This should include:

- Instructions on how to download, install, and run the application.
- A link to your public code repository (e.g., GitHub), (or Expo Snack link, if applicable).
- Clear guidance on any required tools, dependencies, or configurations.
- A user guide explaining how to use the main features of the application.

Additionally, your Git repository must include a **README.md** file containing:

- Installation instructions.
- Usage guidelines.
- Any additional setup notes required to run the project successfully.

5. Critical Thinking, Conclusion & Recommendations (Max. 5 Marks)

The final section should demonstrate **reflection and critical thinking**.

You should:

- Summarise the overall outcome of the project.
- Reflect on challenges encountered and how they were addressed.
- Identify lessons learned during development.
- Provide recommendations for future improvements or extensions to the application.

This section should show evidence of **evaluative thinking**, not just description.

*****All sections will be evaluated based the overall quality, appearance, and the level of screenshots and descriptions of key parts of your report, ensuring clarity and enhancing the visual appeal of your documentation.*

Assessment Breakdown : Part B - Practical Component – Technical Demonstration (60%)

Assessed based on your application's design, functionality, and incorporation of the minimum required features, as well as any additional features you include. However, **the ultimate evaluation and the marks you earn will depend on your demonstrated understanding of these technical achievements**. You must confidently navigate through your code, explain your implementation, and answer specific questions about your approach. This requirement applies to each section being evaluated in this component.

1. Deployment & Demonstration (10 Marks):

A fully functional and properly deployed mobile application is essential. Install your app on a physical device, ensuring that it runs seamlessly. Multiple valid deployment options, including an installable Progressive Web App (PWA) with a Fully Qualified Domain Name (FQDN), a Native App via Expo Go, or a Native App (.apk) installed on an Android device. Hosting your app using platforms such as Vercel or Netlify is also allowed. The demonstration must be well-prepared, clearly showcasing your app's features, design, and functionality. During your demo, you should be able to navigate your app smoothly, provide explanations of key functionalities, and respond to technical questions confidently.

2. App Design – Functionality & User Experience (UI/UX) (10 Marks):

Your application should be well-designed, ensuring a professional and visually appealing user interface. A strong focus should be placed on creating an intuitive, accessible, and engaging experience. The design should be responsive, optimised for different screen sizes and devices, and should enhance the user's interaction with the application. A well-thought-out layout, effective use of colours, typography, and navigation structures will contribute to higher marks. Originality in design and a unique approach to UI/UX elements will also be taken into consideration.

3. On-Device Storage & CRUD (15 Marks)

Your project must demonstrate persistent on-device storage, enabling users to create, retrieve/read, update, and delete (CRUD) data. on-device storage be demonstrated through various methods such as localStorage, IndexedDB (via Dexie or similar libraries), or AsyncStorage for React Native applications. The implementation should ensure that data is stored reliably and persists even after app restarts. Additionally, the CRUD functionalities should be well-integrated, allowing users to interact with and manage data smoothly. Your explanation during the demonstration should highlight how your chosen storage solution works and why it was selected for your application. Well-structured, and optimised storage solutions will contribute to a stronger evaluation.

4. Device APIs (15 Marks)

Your mobile application must incorporate at least two hardware APIs to enhance functionality and provide real-world usability. Common examples include geolocation for location-based services, camera for capturing images or scanning QR codes, and SMS or contact APIs for communication features. You may also explore other hardware APIs to demonstrate initiative and creativity. The assessment will focus on how well these APIs are implemented, their effectiveness within the app, and how they contribute to solving a specific problem. During the demonstration, you should be able to explain how the APIs were integrated, provide a walkthrough of their usage, and showcase practical examples of their functionality.

5. Code Quality & Extras (10 Marks)

The quality of your code plays a significant role in this evaluation. Your implementation should demonstrate clean, readable, and well-structured code that follows best practices in React

development. Higher marks will be awarded for projects that showcase originality in coding, effective use of React features such as hooks, state management, and component architecture, and seamless integration of external libraries or tools. You are also encouraged to include additional functionalities beyond the module requirements, such as offline content, advanced API integrations (e.g., Firebase, Redux), cloud connectivity using frameworks like Laravel or the MERN stack, or any other enhancements that add value to your application. Your ability to articulate and justify these extra features during your demonstration will contribute to the final assessment.
IMPORTANT: *Simply including all required sections is not sufficient to achieve a high grade. Each section will be evaluated on the depth of analysis, the clarity of reasoning, and how well your choices are justified in relation to the specific case study. Marks will be awarded for critical thinking, quality of argument, professional presentation, and the effective use of visuals (e.g., diagrams, tables, screenshots) that enhance understanding and clarity.*

Part A Written Component (Rubric)	**Below Passing Threshold (<40%)**	**Basic competence** **(40%-49%)**	**Good** **(50%-59%)**	**Very Good** **(60%-69%)**	**Excellent** **(70%-100%)**
Presentation & Visual Appeal (max. 5 Marks) (<40%) 0-1 Marks (40%-49%) 2 Marks (50%-59%) 3 Marks (60%-69%) 4 Marks (70%-100%) 5 Marks	Poorly presented document, lacks organisation, and contains no visual aids or examples.	Basic presentation, minimal use of screenshots or visuals, lacks polish.	Well-organised document with effective use of visuals such as screenshots and code snippets.	Highly professional layout with excellent visuals, clear formatting, and logical organisation.	Exceptional presentation with a creative and professional design, high-quality visuals, and excellent attention to detail.
Introduction & Feature Explanation (max. 10 Marks) (<40%) 0-3 Marks (40%-49%) 4 Marks (50%-59%) 5-6 Marks (60%-69%) 7 Marks (70%-100%) 8-10 Marks	Lacks a clear introduction or fails to describe the chosen feature. Explanation is minimal or incorrect.	Basic introduction and feature explanation but lacks depth or clarity.	Clear introduction and explanation of the chosen feature with relevant examples, though lacks refinement.	Well-written introduction and detailed explanation of the feature with good examples and relevance to the project.	Excellent introduction, thorough and insightful explanation of the feature, with well-justified relevance to the project.
Technical Understanding & Documentation (10 Marks) (<40%) 0-3 Marks (40%-49%) 4 Marks (50%-59%) 5-6 Marks (60%-69%) 7 Marks (70%-100%) 8-10 Marks	Fails to demonstrate an understanding of the chosen technical feature or its relevance to the presented project. No clear explanation of how the feature was implemented.	Basic understanding of the technical feature with limited clarity. Explanation of the project implementation is vague or incomplete.	Demonstrates a reasonable understanding of the chosen technical feature and provides a clear explanation of its implementation in the project. Shows how it contributed to solving a specific problem but lacks depth.	Strong understanding of the technical feature and its role within the project. Provides a detailed explanation of the implementation and clearly connects the feature to solving a specific problem in the project.	Exceptional understanding of the technical feature, with a thorough and insightful explanation of its implementation in the project. Clearly and convincingly demonstrates how the feature helped solve a specific problem, supported by detailed examples and evidence.
Project Setup & User Guide (10 Marks) (<40%) 0-3 Marks (40%-49%) 4 Marks (50%-59%) 5-6 Marks (60%-69%) 7 Marks (70%-100%) 8-10 Marks	Missing or unclear instructions for downloading, installing, and testing.	Instructions provided but are incomplete or not user-friendly.	Clear and mostly complete instructions for downloading, installing, and testing the application.	Comprehensive guide with clear, user-friendly steps, including well-chosen screenshots and code extracts.	Flawless, visually appealing guide with excellent formatting, screenshots, and code snippets enhancing understanding.
Critical Thinking, Conclusion & Recommendations (5 Marks) (<40%) 0-1 Marks (40%-49%) 2 Marks (50%-59%) 3 Marks (60%-69%) 4 Marks (70%-100%) 5 Marks	Fails to demonstrate critical thinking, lacks meaningful conclusions, and provides no recommendations.	Basic conclusions drawn with limited evidence of critical thinking. Recommendations are vague or incomplete.	Reasonable conclusions based on evidence, with practical recommendations that address some project aspects.	Strong conclusions supported by critical analysis, with well-thought-out, actionable recommendations.	Insightful and impactful conclusions drawn from critical analysis, accompanied by innovative and realistic recommendations.

Part B Practical Component (Rubric)	**Below Passing Threshold (<40%)**	**Basic competence (40%-49%)**	**Good (50%-59%)**	**Very Good (60%-69%)**	**Excellent (70%-100%)**
Deployment & Demonstration (10 Marks): (<40%) 0-3 Marks (40%-49%) 4 Marks (50%-59%) 5-6 Marks (60%-69%) 7 Marks (70%-100%) 8-10 Marks	The application is not functional or not properly installed. The demonstration is poorly prepared and ineffective.	The application is deployed but crashes when open it or proper demonstration doesn't cover main points.	The application is functional and installed correctly, with a reasonable demonstration of its features.	The application is well-deployed, and the demonstration is structured and confidently delivered.	The application is flawlessly deployed, and the demonstration is engaging, professional, and thorough.
App Design – Functionality & User Experience (UI/UX) (10 Marks): (<40%) 0-3 Marks (40%-49%) 4 Marks (50%-59%) 5-6 Marks (60%-69%) 7 Marks (70%-100%) 8-10 Marks	The design lack's structure, creativity, and does not adhere to basic UI/UX principles.	The design is basic with limited effort toward user experience and aesthetics.	The app has a functional and reasonably good UI/UX, with a clear structure and layout.	The app demonstrates polished UI/UX with an innovative approach to enhancing user interactions.	The app features exceptional UI/UX design, with a highly professional and creative interface that is unique and user centred.
On-Device Storage & CRUD (15 Marks) (<40%) 0-5 Marks (40%-49%) 6-7 Marks (50%-59%) 8-9 Marks (60%-69%) 10-11 Marks (70%-100%) 12-15 Marks	No meaningful storage implementation or CRUD functionality is provided.	Basic CRUD functionality is implemented, but persistence is minimal or poorly integrated (or vice versa).	A reasonable CRUD implementation is provided with structured storage solutions.	The application features strong on-device storage integration and efficient CRUD operations.	The application showcases sophisticated implementation, seamless data persistence, and optimised CRUD operations, supported by clear explanations and examples.
Device APIs (15 Marks) (<40%) 0-5 Marks (40%-49%) 6-7 Marks (50%-59%) 8-9 Marks (60%-69%) 10-11 Marks (70%-100%) 12-15 Marks	Minimal or no meaningful integration of hardware APIs.	Basic API implementation is provided but lacks functionality or relevance. At least one hardware API.	APIs are well-integrated with functional examples. Requires two hardware APIs.	APIs are strongly implemented, with clear, relevant use cases and examples. Requires two hardware APIs.	APIs are innovatively and practically integrated, contributing significantly to the application's functionality.
Code Quality & Extras (10 Marks) (<40%) 0-3 Marks (40%-49%) 4 Marks (50%-59%) 5-6 Marks (60%-69%) 7 Marks (70%-100%) 8-10 Marks	The code is poorly written, disorganised, and lacks originality or meaningful functionality.	The code follows basic standards, with minor customisation or limited feature integration.	The code is well-structured and includes some advanced integrations.	The code demonstrates strong practices and effective integration of extra features.	The code is of high quality, original, and includes advanced features and outstanding integrations.